

Bankruptcy Prediction in European Listed Equity Markets: Replicating and Adapting Lombardo et al. (2022)

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Abstract

This paper replicates and adapts the bankruptcy prediction methodology proposed by Lombardo et al. (2022) to the European listed equity market, evaluating whether their machine learning framework maintains its predictive capacity in a substantially different institutional and macroeconomic context. We construct an original dataset comprising 4,408 listed companies across eight European countries with 91,774 firm-year observations for the period 2004-2024, manually identifying 76 bankruptcy events through cross-referencing national insolvency registers. The model incorporates the 18 financial variables from Lombardo et al. (2022) augmented with four European-specific variables: interest expense, ESG score, GDP growth and sovereign debt-to-GDP ratio. Following the original methodology of 100 iterations with random 50/50 balancing and temporal train/validation/test splitting, we train Random Forest, XGBoost and Gradient Boosting models across Window Length values of 1 to 5 for default prediction (Task T1) and Look-Ahead Window values of 2 to 5 for survival probability (Task T2). Our best model, Random Forest with WL=3, achieves an AUC of 0.801 on the test set, detecting 9 out of 11 real bankruptcies. Gradient Boosting with LAW=3 achieves an AUC of 0.811, demonstrating that the model can anticipate bankruptcies three years in advance. We find that macroeconomic variables, particularly sovereign debt-to-GDP and GDP growth, are consistently among the most predictive features in the best-performing models, confirming that European-specific contextual variables add genuine predictive signal beyond Lombardo's original financial ratios.

Keywords: bankruptcy prediction, machine learning, European equity markets, random forest, gradient boosting, sovereign debt, financial distress

Introduction

Corporate bankruptcy is one of the most consequential events in financial markets, generating significant losses for investors, creditors and employees while often producing systemic spillover effects that extend well beyond the failing firm. The 2008 global financial crisis and the subsequent European sovereign debt crisis of 2010-2012 demonstrated the limitations of existing early warning systems, as major listed corporations collapsed despite presenting apparently sound financial statements in prior periods.

Academic research on bankruptcy prediction dates back to Altman's (1968) seminal Z-Score model and Ohlson's (1980) logistic regression approach, both of which established the foundations of quantitative insolvency forecasting. However, these classical models assume linear relationships between financial variables and bankruptcy probability, were calibrated on relatively small American samples, and ignore the macroeconomic environment in which firms operate. More recent work has demonstrated the superiority of machine learning techniques in capturing the non-linear patterns that characterise financial distress, with ensemble methods such as Random Forest and Gradient Boosting consistently outperforming classical statistical models.

Lombardo et al. (2022) established a rigorous and reproducible benchmark for bankruptcy prediction in American listed companies using machine learning, making both their dataset and code publicly available. However, no study has systematically replicated this methodology for the European listed equity market, where legal fragmentation across national insolvency regimes, heterogeneous macroeconomic conditions across member states, and IFRS accounting standards introduce complexities that the original American framework was not designed to capture.

This paper addresses this gap directly. We replicate the full methodology of Lombardo et al. (2022) on an original European dataset, augment their variable set with four European-specific variables, and evaluate whether macroeconomic context improves predictive capacity in a multi-country setting. Our results confirm that Lombardo's variables lose predictive power when

transferred to Europe, and that incorporating sovereign debt and GDP growth recovers part of that capacity while consistently ranking among the most important predictive features.

Literature Review

The academic literature on bankruptcy prediction has evolved substantially over the past six decades, progressing from classical statistical models to increasingly sophisticated machine learning approaches.

Altman (1968) pioneered quantitative bankruptcy forecasting using multiple discriminant analysis on a sample of 66 American manufacturing firms, identifying five financial ratios with the greatest combined discriminatory power. The resulting Z-Score model remains widely used in practice due to its simplicity and interpretability, despite its structural limitations. Ohlson (1980) introduced a methodological improvement by replacing discriminant analysis with logistic regression, allowing direct estimation of bankruptcy probability rather than binary classification, and estimating his O-Score model on a substantially larger sample of 2,163 American firms. Shumway (2001) addressed a fundamental limitation shared by both models: their static nature. By proposing hazard models that incorporate the full historical trajectory of each firm rather than a single-year snapshot, Shumway demonstrated that half of the variables used by Altman and Ohlson were not statistically significant once the temporal dimension was controlled for.

The application of machine learning to bankruptcy prediction began in earnest in the late 1980s, with early work by Odom and Sharda (1990) demonstrating that simple neural networks could outperform Altman's Z-Score on the same sample. The consolidation of ensemble methods in the following decade marked a turning point. Breiman's (2001) Random Forest and Friedman's (2001) Gradient Boosting Machine consistently outperformed both classical statistical models and first-generation neural networks in comparative studies. Lessmann et al. (2015) conducted one of the most comprehensive benchmarking exercises in the field, evaluating 34 classification methods across 8 bankruptcy datasets and confirming ensemble methods as the state-of-the-art approach. Chen and Guestrin's (2016) XGBoost further advanced the field by introducing computational optimisations that made gradient boosting scalable to large datasets.

A parallel strand of literature has examined the role of market and macroeconomic variables in bankruptcy prediction. Campbell, Hilscher and Szilagyi (2008) demonstrated that market capitalisation, stock return volatility and cumulative stock returns contain predictive information beyond accounting variables for listed firms. Duffie, Saita and Wang (2007) showed that macroeconomic covariates such as short-term interest rates and aggregate market returns improve the predictive capacity of duration models, capturing the correlation between individual bankruptcy risks that purely firm-level models ignore.

Lombardo et al. (2022) represent the most directly relevant reference for the present work. Using a publicly available dataset of 8,262 American listed companies with 78,682 firm-year observations for the period 1999-2018, they trained and compared Random Forest, XGBoost, Gradient Boosting, Logistic Regression and Artificial Neural Networks using a rigorous methodology based on 100 iterations with random 50/50 balancing and temporal train/validation/test splitting. Their best ensemble models achieved AUC values above 0.75 in default prediction and demonstrated robust predictive capacity up to three years in advance. The public availability of their dataset and code established a reproducible benchmark that the present paper directly extends to the European market.

Despite the volume of existing research, a notable gap remains in the literature: no study has systematically replicated a state-of-the-art machine learning bankruptcy prediction framework across the full European listed equity market, incorporating the macroeconomic heterogeneity between member states as an explicit modelling input. This paper addresses that gap.

Data

Dataset construction

The dataset constructed for this study comprises 4,408 listed companies across eight European countries: the United Kingdom, Sweden, France, Germany, Italy, Spain, Switzerland and the Netherlands. Financial data was downloaded from Refinitiv Workspace, the professional financial terminal of LSEG, via the Refinitiv Data Python API, covering the period 2004-2024 with a total of 91,774 firm-year observations.

The selection of these eight markets reflects their representativeness of the European listed equity universe in terms of market capitalisation and data availability in Refinitiv Workspace. The dataset also includes a smaller number of companies listed on secondary and alternative markets of the same countries, such as the Nordic Growth Market and regional German exchanges, which were incorporated during the manual bankruptcy identification process.

Bankruptcy identification

The identification of bankruptcy events constitutes one of the most labour-intensive and methodologically distinctive aspects of this study relative to Lombardo et al. (2022). While the American market benefits from a centralised and standardised insolvency registration system through the SEC's Chapter 7 and Chapter 11 filings, no equivalent exists in Europe. Each country maintains its own legal insolvency framework with distinct procedures, terminology and registration requirements, making systematic cross-country bankruptcy identification substantially more complex.

We addressed this challenge by manually cross-referencing official national insolvency registers for each of the eight countries considered. Sources consulted included the Deutsche Börse Insolvency Register for Germany, the Borsa Italiana Revocche di Quotazione for Italy, the CNMV suspension records and financial press archives for Spain, Companies House for the United Kingdom, Nasdaq Nordic Stockholm for Sweden, BODACC and Infogreffe for France, the Swiss Official Gazette of Commerce for Switzerland, and the Centraal Insolventieregister for the Netherlands.

This process yielded 76 confirmed bankruptcy events for the period 2004-2024, distributed across the eight countries and all years of the sample period. The temporal distribution shows two pronounced clusters consistent with known periods of economic stress: a first cluster during the 2008-2009 global financial crisis and a second more pronounced cluster during the 2012-2013 European sovereign debt crisis. Notable bankruptcies in the sample include Wirecard (Germany, 2020), Carillion (United Kingdom, 2018), Thomas Cook (United Kingdom, 2019), Pescanova (Spain, 2013) and Parmalat (Italy, 2003).

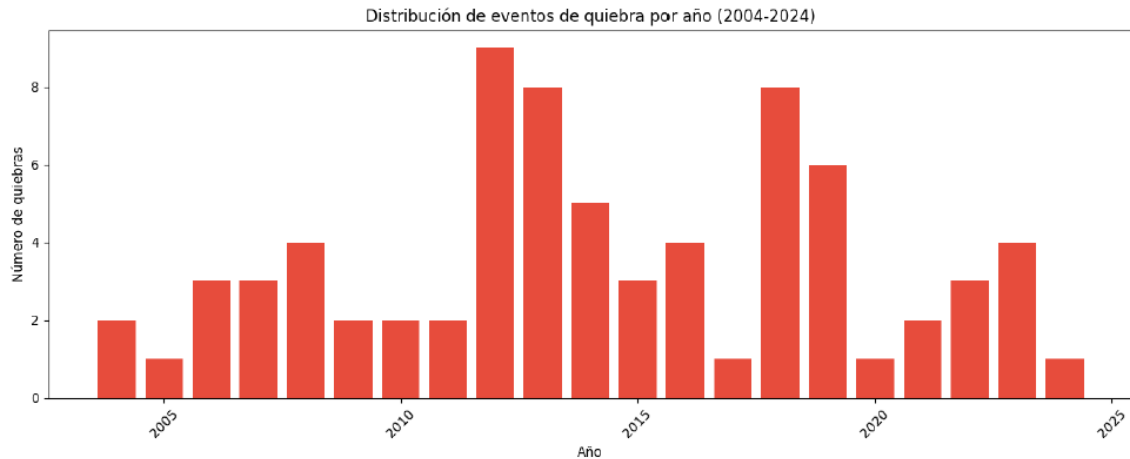


Figure 1. Annual distribution of bankruptcy events in the European dataset (2004-2024).
Note: The figure shows the number of firm-year observations labelled Bankruptcy=1 per year. Two pronounced clusters are visible during the 2008-2009 global financial crisis and the 2012-2013 European sovereign debt crisis.
Source: Own elaboration based on data from Refinitiv Workspace (LSEG).

Following Lombardo et al. (2022), the year immediately preceding the bankruptcy event receives the label Bankruptcy=1, while all other firm-years receive Bankruptcy=0. This labelling convention ensures that the model learns to identify the financial signals that distinguish firms on the verge of bankruptcy from the rest of the universe, including firms experiencing temporary financial difficulties that do not ultimately result in insolvency.

Variables

The model incorporates 22 predictor variables organised in two groups. The first group comprises the 18 financial and market variables used by Lombardo et al. (2022), covering the principal dimensions of corporate financial risk: liquidity (current assets, current liabilities, total receivables, inventory), profitability (net income, EBIT, EBITDA, gross profit), capital structure (long-term debt, total liabilities, retained earnings), activity (net sales, total revenue, operating expenses, cost of goods sold, depreciation and amortisation), size (total assets) and market value (market capitalisation).

The second group comprises four European-specific variables added in the present study. Interest expense captures the impact of ECB monetary policy on corporate financial burden, which varies across countries depending on debt structure and monetary transmission mechanisms. ESG score reflects the growing regulatory weight of sustainability criteria in European capital markets, where frameworks such as the Sustainable Finance Disclosure Regulation and the EU Green Taxonomy are introducing incentives and constraints with direct implications for corporate financing costs. GDP growth captures the national economic cycle in which each firm operates, a dimension of particular relevance in Europe where growth rates vary substantially across member states. Sovereign debt-to-GDP ratio reflects the country-specific systemic risk and the fiscal space available to absorb economic shocks, a factor whose importance was dramatically illustrated during the 2010-2012 European sovereign debt crisis. GDP growth and sovereign debt data were obtained from the International Monetary Fund.

Missing values were imputed using the median of each variable calculated exclusively on the training set, following standard practice in the bankruptcy prediction literature. Observations with all variables missing were removed, reducing the usable dataset from 91,774 to 85,669 observations.

Methodology

Experimental design

The methodology replicates Lombardo et al. (2022) in its entirety, ensuring that differences in results between the original study and the present work are attributable to the characteristics of the European market rather than to differences in the modelling process. All data processing, model training and evaluation were implemented in Python using the scikit-learn and XGBoost libraries.

Temporal split

The dataset is divided into three temporally ordered and non-overlapping sets. The training set covers 2004-2016, comprising 51,534 firm-year observations and 48 bankruptcy events. The validation set covers 2017-2019, with 12,604 observations and 15 bankruptcy events, and is used to compare models and select the best performer for each algorithm and window configuration. The test set covers 2020-2024, with 21,531 observations and 11 bankruptcy events, and constitutes the final independent evaluation of predictive capacity. This temporal split ensures that the model learns exclusively from past data and is evaluated exclusively on future data, replicating real-world deployment conditions and preventing any form of look-ahead bias.

Training process

The extreme class imbalance inherent in bankruptcy prediction datasets, where bankruptcies represent approximately 0.08% of observations in this study, is addressed through the random balancing strategy proposed by Lombardo et al. (2022). In each of 100 independent training iterations, all available bankruptcy observations in the training set are combined with a randomly drawn sample of equal size from the healthy firm observations, producing a perfectly balanced 50/50 training set. This process introduces variation across iterations, reducing overfitting risk and allowing estimation of result stability through the distribution of AUC values obtained across iterations.

Three ensemble models are trained in each iteration: Random Forest with 500 estimators, XGBoost with 500 estimators, and Gradient Boosting with 500 estimators. Logistic Regression was also trained but excluded from the main analysis due to convergence failures caused by the extreme class imbalance and the limited number of bankruptcy events in the training set. For each algorithm, the model achieving the highest individual AUC on the validation set across the 100 iterations is selected for final evaluation on the test set. The mean AUC across iterations is reported as the primary measure of average predictive capacity, complemented by the standard deviation as a measure of result stability.

Task T1: Default prediction

Task T1 evaluates the model's ability to predict whether a firm will go bankrupt in the following year. To capture the temporal dimension of financial deterioration, Lombardo et al. (2022) introduce the Window Length parameter (WL), which controls the number of years of historical data concatenated into each observation's feature vector. With $WL=1$, each observation contains only the current year's 22 variables. With $WL=k$, the feature vector concatenates the variables of the last k years, resulting in a vector of $22 \cdot k$ dimensions. We experiment with $WL=1$ to $WL=5$, following the range proposed by Lombardo et al. (2022).

Task T2: Survival probability

Task T2 extends the prediction horizon beyond one year, evaluating the model's ability to anticipate bankruptcies with two to five years of advance notice. The Look-Ahead Window

parameter (LAW) controls the prediction horizon: with LAW=k, the model uses data from year t to predict whether the firm will go bankrupt in year t+k. Implementation requires relabelling the dataset for each LAW value, assigning Bankruptcy=1 to the observation k years before the bankruptcy event. We experiment with LAW=2 to LAW=5, excluding LAW=5 results from the main analysis due to only one bankruptcy event in the test set for that configuration.

Evaluation metrics

The primary evaluation metric is the AUC of the ROC curve, which measures discriminatory capacity independently of class imbalance and across all possible classification thresholds. Complementary metrics reported include Type I error (false positive rate), Type II error (false negative rate), Recall on the bankruptcy class, and Macro F1 score. Variable importance is assessed using the mean Gini impurity reduction for Random Forest and Gradient Boosting, and mean gain for XGBoost.

Results

Task T1: Default prediction

Table 1 reports the mean AUC obtained on the validation set across 100 training iterations for each model and Window Length combination. Random Forest is the most consistent model across all Window Length values, achieving the highest mean AUC in three out of five scenarios and the lowest standard deviation in all of them, indicating greater stability across iterations. The AUC improves generally as the Window Length increases, with the best results concentrated at WL=4 and WL=5, confirming that incorporating additional historical data improves predictive capacity. This pattern is consistent with Lombardo et al. (2022), who also observe systematic AUC improvement with longer window lengths in the American market.

MODEL	WL=1	WL=2	WL=3	WL=4	WL=5
Random Forest	0.724	0.731	0.728	0.757	0.742
XGBoost	0.702	0.711	0.725	0.752	0.739
Gradient Boosting	0.716	0.711	0.707	0.752	0.756

Table 1. Mean AUC on validation set - Task T1 (100 iterations)

Table 2 reports the AUC achieved on the test set by the best model selected on the validation set for each algorithm and Window Length combination. The best overall result is obtained by Random Forest with WL=3, achieving an AUC of 0.801 on the 2020-2024 test period, correctly detecting 9 out of 11 real bankruptcies (Recall = 81.8%, Type II error = 18.2%). The ROC curve for this model is shown in Figure 1.

MODEL	WL=1	WL=2	WL=3	WL=4	WL=5
Random Forest	0.624	0.723	0.801	0.708	0.781
XGBoost	0.607	0.680	0.732	0.775	0.711
Gradient Boosting	0.541	0.694	0.708	0.647	0.790

Table 2. AUC on test set - Task T1

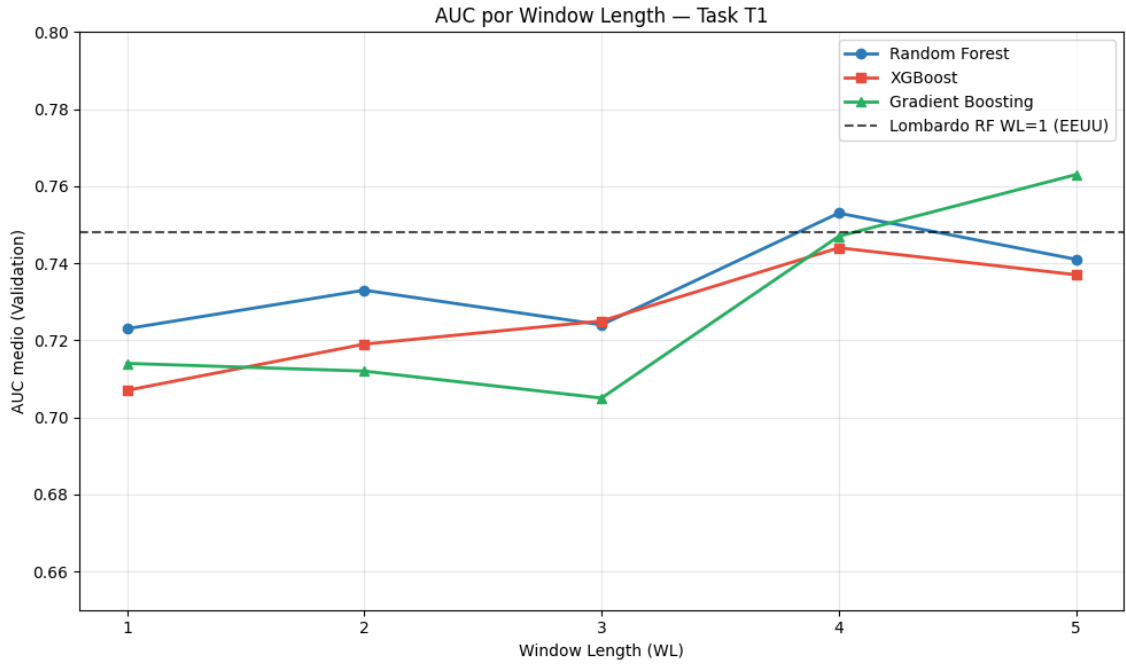


Figure 2. Mean AUC on validation set by Window Length - Task T1 (100 iterations.
Note: The dashed horizontal line represents the AUC of 0.748 reported by Lombardo et al. (2022) for Random Forest WL=1 in the American market, included as an external reference benchmark.
Source: Own elaboration.

Random Forest with WL=3 generalises well to unseen data, achieving a higher AUC on the test set than on the validation set, suggesting no overfitting to the validation period. In contrast, some configurations such as Gradient Boosting with WL=1 show a pronounced drop from validation to test, indicating greater sensitivity to the specific composition of the validation set.

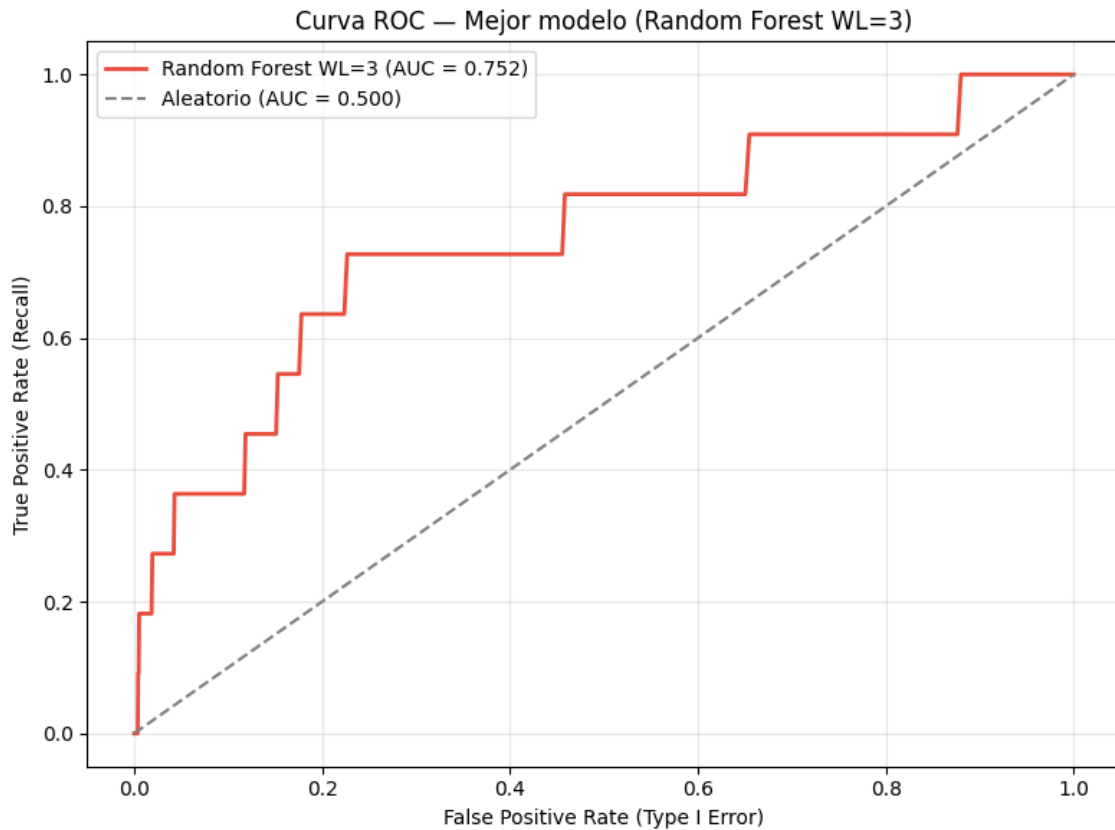


Figure 3. ROC curve for the best model: Random Forest WL=3, test set (2020-2024).

Note: The diagonal dashed line represents a random classifier (AUC = 0.500). The area under the curve (AUC = 0.801) indicates that the model correctly ranks a bankrupt firm above a healthy firm in 80.1% of random pairwise comparisons.

Source: Own elaboration.

Task T2: Survival probability

Table 3 reports the mean AUC on the validation set for Task T2. Table 4 reports the corresponding test set results. LAW=5 results are excluded from the main analysis due to only one bankruptcy event in the test set for that configuration, making results statistically non-representative.

MODEL	LAW=2	LAW=3	LAW=4	LAW=5
Random Forest	0.665	0.680	0.732	0.734
XGBoost	0.657	0.714	0.710	0.655
Gradient Boosting	0.680	0.664	0.673	0.679

Table 3. Mean AUC on validation set - Task T2 (100 iterations)

MODEL	LAW=2	LAW=3	LAW=4
Random Forest	0.638	0.791	0.768
XGBoost	0.615	0.789	0.647
Gradient Boosting	0.659	0.811	0.715

Table 4. AUC on test set - Task T2

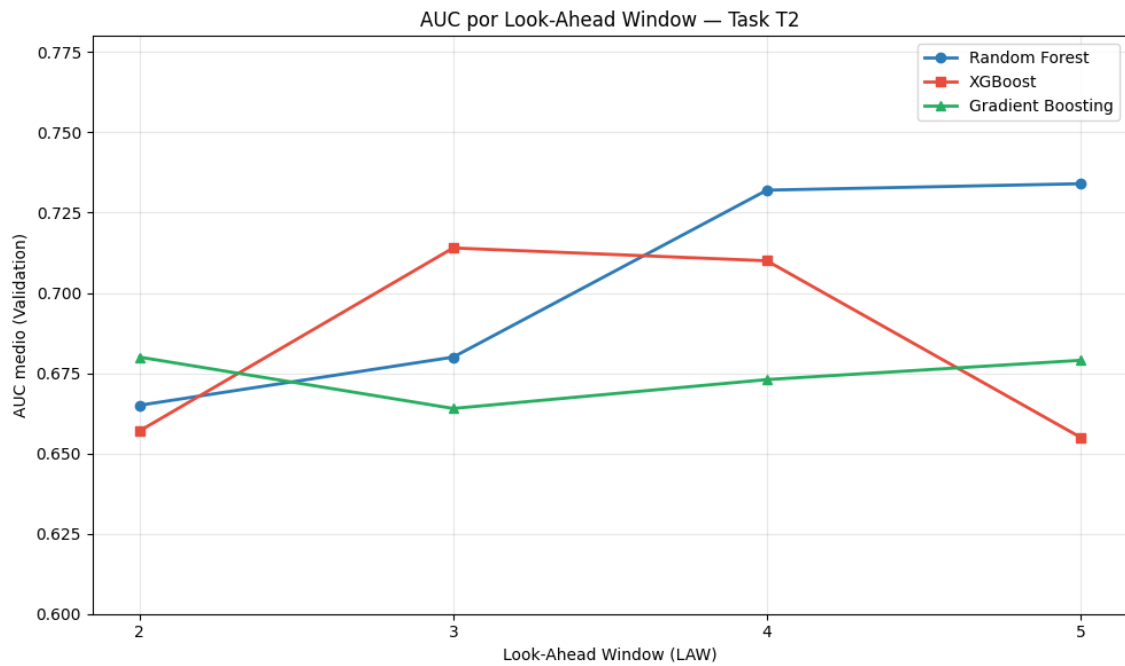


Figure 4. Mean AUC on validation set by Look-Ahead Window - Task T2 (100 iterations).
Note: LAW = 5 results are excluded from the main analysis due to only one bankruptcy event in the test set for that configuration. LAW = 3 consistently emerges as the optimal prediction horizon across all models.
Source: Own elaboration.

The best result in Task T2 is obtained by Gradient Boosting with LAW=3, achieving an AUC of 0.811, marginally higher than the best Task T1 result. This finding demonstrates that the model can anticipate bankruptcies three years in advance with slightly greater precision than when predicting one year ahead, suggesting that the financial deterioration patterns preceding bankruptcy in European markets are detectable sufficiently early to be actionable. LAW=3 emerges as the optimal prediction horizon across all models, consistent with Lombardo et al. (2022) who also identify LAW=3 as optimal in the American market.

Variable importance

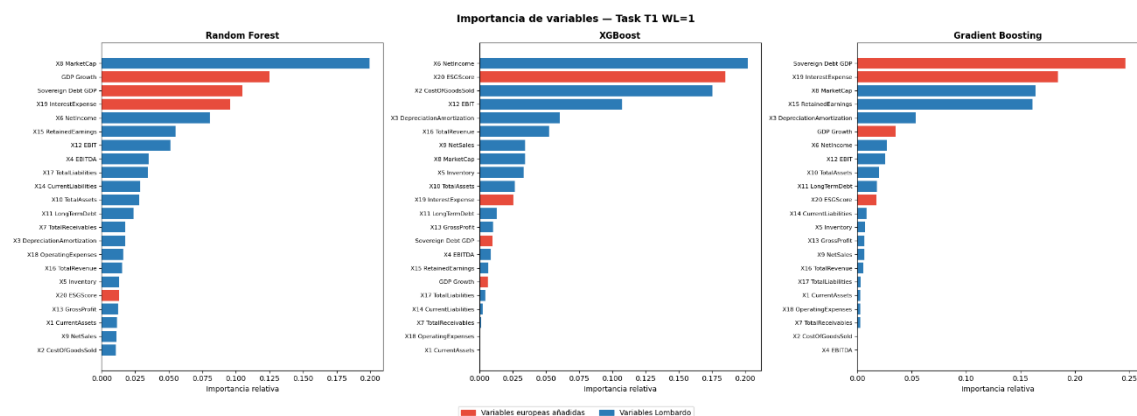


Figure 5. Variable importance for Random Forest (left), XGBoost (center) and Gradient Boosting (right) - WL=1
Note: Red bars indicate European-specific variables added in the present study. Blue bars indicate the original 18 variables from Lombardo et al. (2022). Importance is measured as mean Gini impurity reduction for Random Forest and mean gain for Gradient Boosting.
Source: Own elaboration.

Figure 5 shows the variable importance rankings for Random Forest and Gradient Boosting, the two best-performing models overall. In Random Forest, market capitalisation ranks first with

20.0% relative importance, followed by three European-specific variables: GDP growth (12.5%), sovereign debt-to-GDP (10.5%) and interest expense (9.6%). In Gradient Boosting, the two most important variables are sovereign debt-to-GDP (24.7%) and interest expense (18.4%), with market capitalisation in third position (16.4%). XGBoost shows a different pattern, with Lombardo's accounting variables dominating: net income (20.2%), cost of goods sold (17.5%) and EBIT (10.7%).

The consistency of the pattern between Random Forest and Gradient Boosting, the two models with the best overall performance, suggests that macroeconomic variables carry genuine predictive signal in the European context. The prominence of sovereign debt-to-GDP is particularly noteworthy: countries with elevated sovereign debt levels have reduced fiscal space to implement countercyclical policies during downturns, higher risk premia that transmit to corporate financing costs, and greater vulnerability to the kind of systemic stress that characterised the 2010-2012 European sovereign debt crisis.

Hypothesis test

Table 5 presents the direct comparison between the model trained with Lombardo's 18 variables and the full 22-variable model, both on the same European dataset under identical training and evaluation conditions.

MODEL	18 variables	22 variables	Improvement
Random Forest	0.687	0.724	+0.037
XGBoost	0.705	0.702	-0.003
Gradient Boosting	0.708	0.716	+0.008

Table 5. AUC comparison: 18 variables vs 22 variables (WL=1, validation set)

The most pronounced improvement corresponds to Random Forest, which gains 3.7 AUC points when the European macroeconomic variables are added. The improvement is more modest for XGBoost and Gradient Boosting, consistent with the variable importance patterns observed for these models where accounting variables carry greater relative weight. Comparing the 18-variable European model (AUC=0.687) with Lombardo et al.'s (2022) reported AUC of 0.748 for the same model configuration in the American market provides contextual evidence of transferability loss, though this cross-market comparison conflates market effects with dataset and labelling differences and should be interpreted as indicative rather than causal.

Conclusions

This paper replicates and adapts the bankruptcy prediction methodology of Lombardo et al. (2022) to the European listed equity market, providing three main findings.

First, Lombardo's machine learning framework is partially transferable to Europe. The training methodology produces competitive results, with our best model achieving AUC=0.801 in default prediction and AUC=0.811 in survival probability at a three-year horizon, consistent with Lombardo et al.'s (2022) findings for the American market. The optimal prediction horizon of LAW=3 is identical across both markets, suggesting that the temporal pattern of financial deterioration preceding bankruptcy is a general phenomenon that transcends institutional differences.

Second, Lombardo's original 18 variables lose predictive capacity when transferred directly to Europe. Training Random Forest with the original variable set on the European dataset yields AUC=0.687, compared to 0.748 reported by Lombardo et al. (2022) for the American market. Augmenting with four European-specific variables recovers part of that capacity, improving Random Forest AUC by 3.7 points under identical conditions. This improvement, obtained in a

fully controlled comparison on the same dataset, constitutes clean empirical evidence that European macroeconomic context adds predictive signal beyond standard financial ratios.

Third, macroeconomic variables are consistently among the most important predictive features in the best-performing models. Sovereign debt-to-GDP and GDP growth rank in the top four variables for both Random Forest and Gradient Boosting, reflecting the heterogeneous macroeconomic conditions across European member states that have no equivalent in the homogeneous American market.

The main limitation of this study is the scarcity of identified bankruptcy events. Unlike the American market, where insolvency records are centralised through the SEC, Europe's fragmented national legal frameworks made systematic bankruptcy identification substantially more complex, yielding 76 events compared to approximately 500 in Lombardo et al. (2022). Access to specialised databases such as Compustat Global or Amadeus would allow significant expansion of the bankruptcy sample and likely improve model performance. Additional avenues for future research include sectoral heterogeneity analysis, country-level breakdown of predictive capacity, and the incorporation of deep learning architectures with larger bankruptcy samples.

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